some cases, as the ticking of our best atomic clocks. Those pulses can *only* be explained as due to beams of radiation shining off a neutron star's surface and swinging past Earth as the star rotates—the analogue of a rotating light beacon at an airport or in a lighthouse. Why is this the only possible explanation? Such precise timing can come only from the rotation of a massive object with massive inertia and thus massive resistance to erratic forces that would make the timing erratic; of all the massive objects ever conceived by the minds of astrophysicists, only neutron stars and black holes can spin at the enormous rates (hundreds of rotations per second) of some pulsars; and only neutron stars, not black holes, can produce rotating beams, because black holes cannot have "hair." (Any source of such a beam, attached to the hole's horizon, would be an example of the type of "hair" that a black hole cannot hang on to.[1]

An unequivocal black-hole signature, analogous to a pulsar's pulses, has been sought by astronomers in Cyg X-1 for twenty years, to no avail. An example of such a signature (suggested in 1972 by Rashid Sunyaev, a member of Zel'dovich's team) is pulsar-like pulses of radiation produced by a swinging beam that originates in a coherent lump of gas orbiting around the hole. If the lump were close to the hole and held itself together for many orbits until it finally began to plunge into the horizon, then the details of its gradually shifting interval between pulses might provide a clear and unambiguous "I am a black hole" signature. Unfortunately, such a signature has never been seen. There seem to be several reasons: (1) The hot, X-ray-emitting gas moves around the black hole so turbulently and chaotically that coherent lumps may hold themselves together for only one or a few orbits, not many. (2) If a few lumps do manage to hold themselves together for a long time and produce a black-hole signature, the turbulent X-rays from the rest of the turbulent gas evidently bury their signature. (3) If Cyg X-1 is indeed a black hole, then mathematical simulations show that most of the X-rays should come from far outside its horizon—from circumferences roughly 10 times critical or more, where there is much more volume from which X-rays can be emitted than near the horizon. At such large distances from the hole, the gravitational predictions of general relativity and Newton's theory of gravity are approximately the same, so if there were pulses from orbiting lumps, they would not carry a strongly definitive black-hole signature.